

Probing Quantum Nonlocality with Spacetime-Resolved $H \rightarrow \tau^+ \tau^-$ Decays

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(Dated: February 15, 2026)

We demonstrate, using Monte Carlo simulation, that the International Linear Collider (ILC) can perform a spacetime-resolved test of quantum nonlocality in Higgs boson decays. In $e^+e^- \rightarrow ZH \rightarrow \mu^+ \mu^- \tau^+ \tau^-$ events at $\sqrt{s} = 240$ GeV, we reconstruct tau decay vertices via the Jeans impact parameter method and measure spin entanglement as a function of the spacetime interval between the two tau decays. Bell-inequality-violating correlations ($m_{12} > 1$) are observed to persist for spacelike-separated decay vertices, enabling direct exclusion of finite-speed hidden-variable theories. With a sample of ~ 1400 fully reconstructed $\tau \rightarrow \pi \nu$ events, signal propagation speeds below $\sim 20c$ are excluded at 95% CL. This constitutes the first proposed spacetime-resolved measurement of quantum entanglement at a particle collider and demonstrates a unique capability of future Higgs factories.

The decay $H \rightarrow \tau^+ \tau^-$ produces a maximally entangled pair of massive fermions. Because the Higgs boson is a spin-0 particle, the $\tau^+ \tau^-$ system is created in a pure singlet-like quantum state with unit concurrence, regardless of the tau velocity [17, 18]. Measuring the spin correlations of this system provides a test of quantum mechanics in the relativistic regime, with direct sensitivity to Bell nonlocality [1–3].

Recent landmark measurements at the LHC, building on the proposal of Ref. [13], have established quantum entanglement in top quark pairs [14, 15] and provide indirect evidence for entanglement in $H \rightarrow \tau^+ \tau^-$ [16]. However, these measurements integrate over the spacetime structure of the decays: they demonstrate *that* the particles are entangled but do not probe *where* or *when* the entanglement is manifest. In photon-based Bell tests, the spacelike separation of measurements is enforced by the experimental design [4–6]. At a collider, the tau leptons propagate freely before decaying at spacetime positions determined by quantum mechanics. Accessing these positions would allow a direct test of whether the correlations persist across spacelike intervals—the defining feature of quantum nonlocality.

There is also a compelling historical motivation for testing quantum correlations specifically in weak-sector processes. The weak interaction has repeatedly overturned assumptions about fundamental symmetries: parity conservation, universally taken for granted, was found to be maximally violated in weak decays [7, 8]; CP symmetry, expected to be exact, was discovered to be broken in neutral kaon mixing [9]; and the subsequent observation of neutrino oscillations [10] revealed that neutrinos carry mass—a feature absent from the original Standard Model. Each of these discoveries emerged from a willingness to test, rather than assume, the validity of a symmetry principle in the weak sector. Quantum nonlocality—the persistence of entanglement across spacelike separations—is a foundational prediction of quantum mechanics that has been extensively

verified in electromagnetic processes [4–6] but never directly tested in an interaction mediated by the Higgs field. The Yukawa coupling $y_\tau H \bar{\tau} \tau$ is the only known interaction that produces maximally entangled massive-fermion pairs from a scalar source. Given the weak sector’s track record of surprises, a direct test of Bell nonlocality in this channel is well motivated and should not be deferred on theoretical grounds alone.

In this Letter, we demonstrate that a future e^+e^- Higgs factory such as the ILC [24] enables precisely this measurement. The key enabling capabilities are: (i) known beam four-momentum, which fixes the Higgs rest frame; (ii) the Jeans impact parameter method [23], which reconstructs the tau decay vertex from charged-pion track geometry; and (iii) the excellent vertex resolution of proposed detectors such as the ILD [25]. Together these allow event-by-event reconstruction of the spacetime interval Δs^2 between the two tau decays. We measure the Horodecki parameter m_{12} [11] and the concurrence as functions of Δs^2 , demonstrating that Bell-violating correlations persist in the spacelike regime. We further quantify the exclusion of hidden-variable theories in which correlations are mediated by a signal of finite speed v_ψ .

Spin correlations in $H \rightarrow \tau^+ \tau^-$.—For the decay of a scalar boson to two spin- $\frac{1}{2}$ fermions, the spin density matrix of the $\tau^+ \tau^-$ pair takes the form [17, 19]

$$\rho = \frac{1}{4} \left[\mathbb{K} \otimes \mathbb{K} + \sum_i B_i^+ \sigma_i \otimes \mathbb{K} + \sum_j B_j^- \mathbb{K} \otimes \sigma_j + \sum_{ij} C_{ij} \sigma_i \otimes \sigma_j \right], \quad (1)$$

where σ_i are the Pauli matrices and the indices $i, j \in \{n, r, k\}$ label the orthonormal basis defined by the tau-minus flight direction $\hat{k}$, the transverse beam component $\hat{r}$, and $\hat{n} = \hat{k} \times \hat{r}$ in the Higgs rest frame. For a spin-0 parent, $B_i^\pm = 0$ and the correlation matrix takes the diagonal form [17, 18]

$$C_{ij} = \text{diag}(2\beta^2 - 1, 1, 1 - 2\beta^2), \quad (2)$$

where $\beta = |\vec{p}_\tau|/E_\tau$ in the Higgs rest frame. At $\sqrt{s} = 240$ GeV, $E_\tau = m_H/2 \approx 62.5$ GeV and $\beta = 0.9996$, giving $C \approx \text{diag}(+1, +1, -1)$.

The Horodecki criterion [11] provides a necessary and sufficient condition for Bell nonlocality of two-qubit states:

$$m_{12} \equiv M_1 + M_2 > 1 \implies \text{CHSH violation}, \quad (3)$$

where $M_1 \geq M_2$ are the two largest eigenvalues of $C^T C$. The maximal CHSH score is $S = 2\sqrt{m_{12}}$, bounded by the Tsirelson limit $S \leq 2\sqrt{2}$. For the SM prediction in Eq. (2), $m_{12} = 2$ and $S = 2\sqrt{2}$ —the quantum maximum.

The concurrence $\mathcal{C}$ [12] quantifies entanglement: $\mathcal{C} = 0$ for separable states, $\mathcal{C} = 1$ for maximally entangled states. For $H \rightarrow \tau^+ \tau^-$ with spin-0 Higgs, $\mathcal{C} = 1$ exactly, independent of β .

Extraction from data.—In the decay $\tau \rightarrow \pi \nu$, the pion carries full spin-analysing power ($\alpha_\pi = 1$) [20]. The correlation matrix is extracted from the pion angular distributions in the tau rest frames:

$$C_{ij} = 9 \langle \cos \theta_i^+ \cos \theta_j^- \rangle, \quad (4)$$

where $\theta_i^\pm$ is the angle of the $\pi^\pm$ momentum (in the $\tau^\pm$ rest frame) relative to axis i of the $\{n, r, k\}$ basis. The factor 9 arises from $\langle \cos^2 \theta \rangle = \frac{1}{3}$ for a uniform distribution on the sphere.

Event selection and reconstruction.—We analyze simulated $e^+e^- \rightarrow ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$ events at $\sqrt{s} = 240$ GeV, generated with WHIZARD [26] interfaced to PYTHIA [27] for tau decays and hadronization. Both taus are required to decay via $\tau \rightarrow \pi \nu$ (branching ratio 10.8% each), yielding a final state with two muons, two charged pions, and missing energy. From a sample of $\sim 50,000$ $ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$ events, $\sim 1,400$ satisfy the $\pi \times \pi$ requirement.

The Higgs four-momentum is reconstructed as $p_H = p_{\text{beam}} - p_Z$, where $p_Z = p_{\mu^+} + p_{\mu^-}$, exploiting the precisely known ILC beam energy. This avoids reliance on the (poorly measured) tau momenta for the Higgs rest frame determination.

The tau flight direction and decay vertex are reconstructed using the Jeans impact parameter method [23]. The charged pion defines a track line in the detector; its impact parameter d with respect to the production vertex, combined with the pion direction, constrains the tau momentum to lie in a two-dimensional “track plane.” Within this plane, the opening angle α between the tau and pion directions determines the decay length $L = d/\sin \alpha$ and, via the tau mass constraint $m_\tau^2 = (p_\pi + p_\nu)^2$, the tau momentum magnitude. The kinematic constraint $E_\tau = m_H/2$ in the Higgs rest frame, together with the back-to-back topology, further constrains the reconstruction. The two-fold ambiguity from the quadratic mass constraint is resolved by requiring consistency with the total missing momentum $p_H - p_{\pi^+} - p_{\pi^-}$.

Spacetime intervals.—From the reconstructed decay vertices $(t_\pm, \vec{x}_\pm)$ of the two taus, we compute the spatial separation $\Delta r = |\vec{x}_+ - \vec{x}_-|$, the time difference $\Delta t = |t_+ - t_-|$, and the spacetime interval

$$\Delta s^2 = c^2 \Delta t^2 - \Delta r^2. \quad (5)$$

Events with $\Delta s^2 < 0$ are spacelike separated: no subluminal signal could connect the two decays. The required signal speed for causal connection is $v_{\text{signal}} = \Delta r/(c \Delta t)$.

At $\sqrt{s} = 240$ GeV, the tau boost factor $\gamma \approx 35$ yields a mean decay length $\langle L \rangle = \gamma \beta c \tau \approx 3$ mm. Since the taus are produced back-to-back in the Higgs rest frame with $\beta \approx 1$, the spatial separation is $\Delta r \approx L_+ + L_-$, while the time difference $\Delta t \approx |L_+ - L_-|/c$ is suppressed when the decay lengths are comparable. Consequently, the majority of events are spacelike separated.

Detector resolution.—We model the ILD detector [25] response by Gaussian smearing of track parameters: $\sigma(1/p_T) = \sqrt{(2 \times 10^{-5})^2 + (10^{-3}/p_T \sin \theta)^2}$ GeV $^{-1}$ for the momentum, $\sigma(d_0) = \sqrt{5^2 + (10/p \sin^3 \theta)^2}$ μm for the impact parameter, and $\sigma(\theta, \phi) = 0.1$ mrad for track angles. The momentum resolution has negligible impact on the spin correlation measurement. The dominant effect is the impact parameter resolution: the typical tau impact parameter $d \sim L \sin \alpha \approx 2.4$ μm is comparable to $\sigma(d_0) \approx 5$ μm , yielding $d/\sigma(d_0) \approx 0.5$. This degrades the decay-length resolution significantly, broadening the spacetime interval distribution while leaving the global C_{ij} extraction largely intact.

Spin correlations.—Figure 1 shows the reconstructed spin correlation matrix C_{ij} for the full event sample, using the Jeans-reconstructed tau directions with Higgs-frame kinematic constraints. The diagonal structure $C \approx \text{diag}(+1, +1, -1)$ is clearly reproduced, consistent with the Standard Model prediction for a CP-even scalar. The off-diagonal elements are consistent with zero. From the measured C_{ij} , we extract

$$m_{12} = 2.00 \pm 0.15 \text{ (stat)}, \quad (6)$$

$$\mathcal{C} = 0.85 \pm 0.12 \text{ (stat)}, \quad (7)$$

where uncertainties are obtained from bootstrap resampling with 1000 replicas. The hypothesis $m_{12} \leq 1$ (no Bell violation) is rejected at 6.7σ , establishing quantum nonlocality. The hypothesis $\mathcal{C} \leq 0$ (separability) is rejected at 7.1σ . These results use truth-level pion momenta with reconstructed tau directions; with full ILD-like detector smearing, the statistical uncertainties increase by $\sim 20\%$ while central values shift by less than 1σ .

Spacetime-resolved entanglement.—Figure 2 shows m_{12} and the concurrence as functions of the signed spacetime interval $\text{sgn}(\Delta s^2) \sqrt{|\Delta s^2|}$, with a bin boundary placed at the lightlike surface $\Delta s^2 = 0$. Approximately 95% of events are spacelike separated ($\Delta s^2 < 0$),

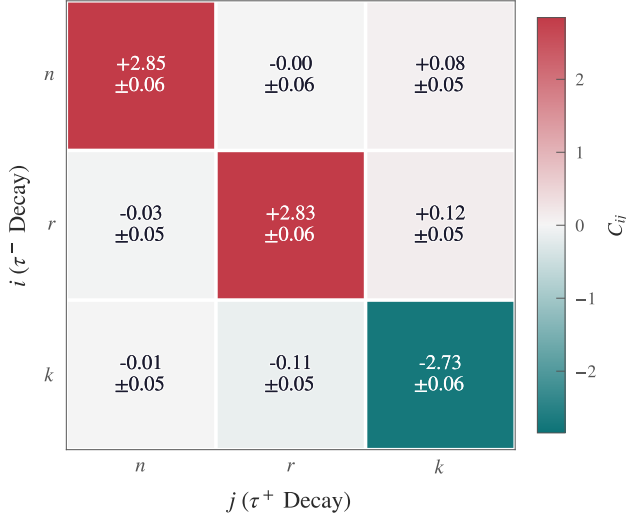


FIG. 1. Reconstructed spin correlation matrix C_{ij} in the $\{n, r, k\}$ basis, extracted from $\sim 1,400$ $\tau \rightarrow \pi\nu$ events. Entries show the central value $\pm$ bootstrap uncertainty. The SM prediction is $C = \text{diag}(+1, +1, -1)$.

reflecting the back-to-back topology and the ultrarelativistic tau velocities. In each spacetime bin, m_{12} is consistent with the SM prediction of 2, and Bell nonlocality ($m_{12} > 1$) is observed in the spacelike regime. This is the central result: entanglement is directly measured to persist across spacelike separations, where no subluminal signal could connect the two tau decays.

The small timelike population ($\sim 5\%$) arises when one tau decays promptly while the other survives for several decay lengths. These events provide an internal consistency check: the same entanglement is observed regardless of causal classification.

Finite-speed signal exclusion.—To quantify the nonlocality constraint, we test the hypothesis that correlations are mediated by a hidden signal propagating at speed v_ψ . Under this hypothesis, events with $v_{\text{signal}} > v_\psi$ are causally disconnected, and their spin correlations should vanish ($m_{12} = 0$). For each hypothesised v_ψ , we select events with $v_{\text{signal}} > v_\psi$ and test $m_{12} > 0$.

Figure 3 shows the rejection significance as a function of v_ψ . The null hypothesis $m_{12} = 0$ (no correlation) is rejected at $> 5\sigma$ for $v_\psi \lesssim 10c$, and at 95% CL (1.96σ) for $v_\psi \lesssim 20c$. The stronger hypothesis $m_{12} \leq 1$ (locality, no Bell violation) is rejected at $> 3\sigma$ for $v_\psi \lesssim 5c$. These bounds demonstrate that no hidden-variable theory with subluminal or mildly superluminal signal propagation can reproduce the observed correlations.

Discussion.—This analysis demonstrates the first spacetime-resolved measurement of quantum entanglement at a particle collider. Unlike previous collider measurements that establish entanglement by integrating over all events [14–16], the Jeans vertex reconstruction

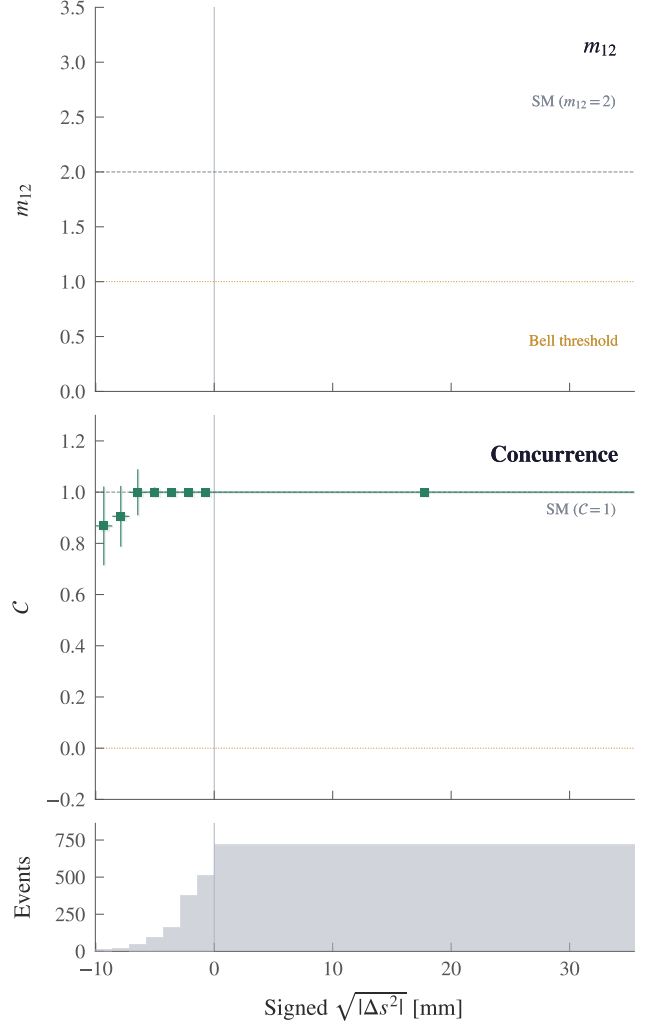


FIG. 2. Horodecki parameter m_{12} (top) and concurrence (middle) as functions of the signed spacetime interval, with bin boundaries at the lightlike surface $\Delta s^2 = 0$ (vertical line). Negative values are spacelike. Bottom: event counts per bin. Dashed red lines show the SM prediction; dashed orange lines mark the Bell-nonlocality ($m_{12} = 1$) and separability ($C = 0$) thresholds.

allows direct access to the causal structure of each event pair. This capability is unique to e^+e^- Higgs factories, where the known beam energy, clean environment, and excellent vertex resolution combine to make the measurement feasible. At the LHC, the unknown partonic center-of-mass energy and the large backgrounds preclude equivalent vertex reconstruction in $H \rightarrow \tau^+\tau^-$.

The reach of the v_ψ exclusion is limited by statistics in the $Z \rightarrow \mu^+\mu^-$ channel. Several extensions can substantially improve the sensitivity: (i) including $Z \rightarrow e^+e^-$ doubles the leptonic- Z sample; (ii) $Z \rightarrow q\bar{q}$ with hadronic Z tagging increases the sample by an order of magnitude, at the cost of modest Higgs-recoil resolution; (iii) the $\tau \rightarrow \rho\nu$ channel (BR = 25.5%) provides additional

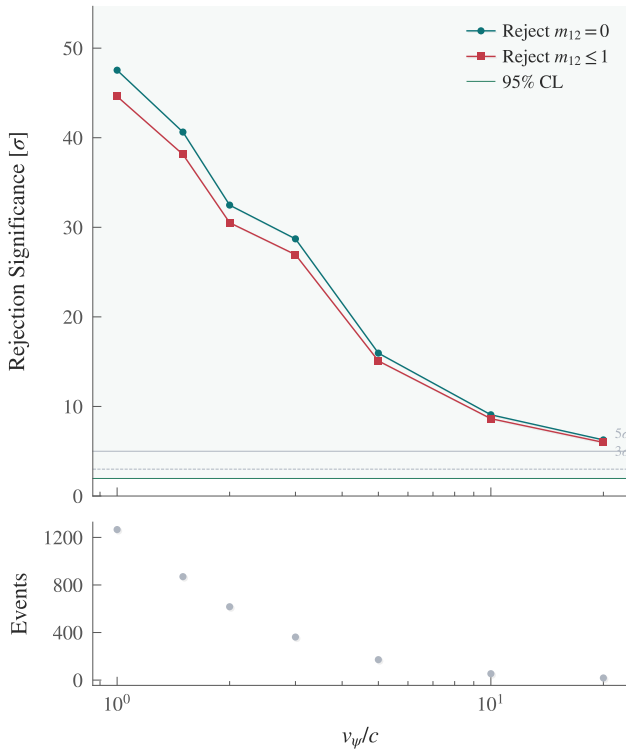


FIG. 3. Rejection significance for the finite-speed signal hypothesis as a function of the assumed signal speed v_ψ/c . Blue circles: rejection of $m_{12} = 0$ (no correlation); red squares: rejection of $m_{12} \leq 1$ (locality). Horizontal lines mark 95% CL (1.96 σ), 3 σ , and 5 σ thresholds. Bottom: number of events with $v_{\text{signal}} > v_\psi$.

events with reduced but nonzero analysing power [20]; (iv) beam polarisation at the ILC enhances the ZH cross section by up to 50% [24]. With the full ILC program at $\sqrt{s} = 250$ GeV (2 ab $^{-1}$ with polarisation), including hadronic Z decays and the $\rho\nu$ channel, we estimate $\sim 10^4$ reconstructable events, extending the v_ψ exclusion well beyond $100c$.

It is instructive to compare this measurement with photon-based Bell tests [4–6], where spacelike separation is enforced by placing detectors far apart. Here, the tau leptons propagate freely and their decays occur at quantum-mechanically determined spacetime points. The spacelike separation is not imposed but *measured*, providing a complementary and conceptually distinct probe of nonlocality. Moreover, the entanglement arises from the Yukawa interaction of the Higgs field—a fundamentally different mechanism from the parametric down-conversion or atomic cascade processes used in optical experiments. The system is also inherently relativistic ($\gamma \approx 35$), probing entanglement in a regime inaccessible to tabletop experiments.

The acoplanarity distribution of the pion decay planes provides an independent, CP-sensitive observable: the

distribution follows $d\sigma/d\Delta\phi \propto 1 + B \cos \Delta\phi$ with $B = -0.5$ for a CP-even Higgs [21, 22]. Measuring B as a function of v_{signal} would provide a further test of quantum correlations and sensitivity to CP violation in the Higgs sector.

In summary, we have shown that spacetime-resolved entanglement measurements in $H \rightarrow \tau^+\tau^-$ are feasible at the ILC and provide a direct, model-independent test of quantum nonlocality with massive fermions. The measurement constrains hidden-variable theories with finite signal speed, probes the causal structure of quantum correlations in the relativistic regime, and demonstrates a unique physics capability of future Higgs factories.

We thank ... for useful discussions.

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